

Assignment 3

Part 1: Mathematical Derivations

Q1. Fitting an SVM by hand

(a)

$$\begin{aligned}\phi(x_1) &= [1, \sqrt{2}x_1, x_1^2]^T = [1, (\sqrt{2})(0), (0)^2]^T = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \\ \phi(x_2) &= [1, \sqrt{2}x_2, x_2^2]^T = [1, (\sqrt{2})(\sqrt{2}), (\sqrt{2})^2]^T = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}\end{aligned}$$

(b)

The vector that connects $\phi(x_1)$ to $\phi(x_2)$ is obtained by

$$\begin{aligned}\phi(x_1) &= w^\perp + \\ \phi(x_1) - \phi(x_2) &= w^\perp \\ w^\perp &= \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix}\end{aligned}\tag{1}$$

thus

$$w = [0, 1, 1]^T\tag{2}$$

is a valid vector that is parallel to optimal w .

(c) margin =

$$\|\phi(x_1) - \phi(x_2)\| = \sqrt{8}\tag{3}$$

which means

$$\frac{1}{\|w\|} = \frac{\sqrt{8}}{2} \implies \|w\| = \frac{1}{\sqrt{2}}\tag{4}$$

(d) note, if $\|w\| = \frac{1}{\sqrt{2}}$, then $w = [0, 0.5, 0.5]^T$ to find bias we can use

$$-1 \left([0, 0.5, 0.5] \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + w_0 \right) = 1\tag{5}$$

the first term is 0, so $w_0 = -1$.

we can check using

$$+1 \left([0, 0.5, 0.5] \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} + w_0 \right) = 1\tag{6}$$

where we get

$$(0 + 1 + 1 + w_0) = 1 \implies w_0 = -1\tag{7}$$

(e)

$$f(\phi(x)) = \text{sign}([0, 0.5, 0.5][1, \sqrt{2}x, x^2] + -1)\tag{8}$$

the decision in the original space is a quadratic decision boundary. since it is one d, it is moreso deciding along roots of some polynomial in higher dimension.

(f)

$$\phi(x_1)^T \phi(x_2) = 1 \tag{9}$$

$$\begin{aligned} K(x_i, x_j) &= (x_i \cdot x_j + 1)^2 \\ K(0, \sqrt{2}) &= (0 \cdot \sqrt{2} + 1)^2 = 1 \end{aligned} \tag{10}$$

and so they do match.

Q2. Deriving the SVM dual

(a)

the lagrangian is written as

$$L(w, b, \alpha) = \frac{1}{2} \|w\|^2 + \sum_i \alpha_i (1 - y_i(w \cdot x_i + b)) \quad (11)$$

(b)

$$\begin{aligned} \frac{\partial L(w, b, \alpha)}{\partial w} &= \frac{\partial}{\partial w} \left(\frac{1}{2} \|w\|^2 + \sum_i \alpha_i (1 - y_i(w \cdot x_i + b)) \right) \\ &= \frac{\partial}{\partial w} \left(\frac{1}{2} \|w\|^2 + \sum_i \alpha_i (y_i(w \cdot x_i)) \right) \\ &= w - \sum_i \alpha_i y_i x_i \end{aligned} \quad (12)$$

$$\begin{aligned} \frac{\partial L}{\partial b} &= \frac{\partial}{\partial b} \left(\frac{1}{2} \|w\|^2 + \sum_i \alpha_i (1 - y_i(w \cdot x_i + b)) \right) \\ &= \frac{\partial}{\partial b} \left(\sum_i \alpha_i (1 - y_i(w \cdot x_i + b)) \right) \\ &= \frac{\partial}{\partial b} \left(- \sum_i \alpha_i y_i b \right) \\ &= - \sum_i \alpha_i y_i \end{aligned} \quad (13)$$

if we set these two equations to zero, we get the values we originally wanted.

w is a vector representing the weights of the examples by eq 12. for each x_i , it is weighted by exactly $\alpha_i y_i$.

(c)

now that we know $w = \sum_i \alpha_i y_i x_i$, substituting back into eq 11

$$w \cdot w = \sum_i \alpha_i y_i x_i \cdot \sum_i \alpha_i y_i x_i = \sum_{i,j} \alpha_i \alpha_j y_i y_j (x_i \cdot x_j) = \sum_{i,j} \alpha_i \alpha_j y_i y_j \delta_{ij} \quad (14)$$

where δ_{ij} is the kronecker delta.

and

$$\sum_i \alpha_i (1 - y_i(w \cdot x_i + b)) = \sum_{ij} \alpha_i \alpha_j y_i y_j (x_i \cdot x_j) + \sum_i \alpha_i y_i b = \sum_{i,j} \alpha_i \alpha_j y_i y_j \delta_{ij} + 0 \quad (15)$$

note that eq 14, eq 15 are actually the same. when we input this back into eq 11,

$$L(\alpha) = \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \delta_{i,j} \quad (16)$$

the restriction conditions are just from the restrictions in the lagrange multipliers ($\alpha_n \geq 0$) and the restriction in the extremum in step b ($\sum_n \alpha_n y_n = 0$)

(d)

the entire model is only dependent on a small portion of the training data. most points are ignored when α_n is zero. in soft margin, α_n is no longer bounded and thus there can exist non margin support vectors (misclassified or insufficiently classified points)

Q3. The kernel trick and valid kernels

(a)

$$\begin{aligned} K(x, z) &= (x_1 z_1)^2 + (x_2 z_2)^2 + (1)^2 + 2(x_1 z_1)(x_2 z_2) + 2(x_1 z_1)(1) + 2(x_2 z_2)(1) \\ &= x_1^2 z_1^2 + x_2^2 z_2^2 + 1 + 2x_1 x_2 z_1 z_2 + 2x_1 z_1 + 2x_2 z_2 \end{aligned} \quad (17)$$

now we can just go entry by entry and define the ϕ map.

$$\phi(x) = \begin{pmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2}x_1 x_2 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ 1 \end{pmatrix} \quad (18)$$

to verify

$$K(x, z) = (x \cdot z + 1)^2 = (5 + 1)^2 = 6^2 = 36 \quad (19)$$

and

$$\begin{aligned} \phi(1, 2) &= \begin{pmatrix} 1^2 \\ 2^2 \\ \sqrt{2}(1)(2) \\ \sqrt{2}(1) \\ \sqrt{2}(2) \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 2\sqrt{2} \\ \sqrt{2} \\ 2\sqrt{2} \\ 1 \end{pmatrix} \\ \phi(3, 1) &= \begin{pmatrix} 3^2 \\ 1^2 \\ \sqrt{2}(3)(1) \\ \sqrt{2}(3) \\ \sqrt{2}(1) \\ 1 \end{pmatrix} = \begin{pmatrix} 9 \\ 1 \\ 3\sqrt{2} \\ 3\sqrt{2} \\ \sqrt{2} \\ 1 \end{pmatrix} \end{aligned} \quad (20)$$

$$\phi(x) \cdot \phi(z) = (1)(9) + (4)(1) + (2\sqrt{2})(3\sqrt{2}) + (\sqrt{2})(3\sqrt{2}) + (2\sqrt{2})(\sqrt{2}) + (1)(1) = 36 \quad (21)$$

(b)

$$G = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{pmatrix} \quad (22)$$

the characteristic polynomial for G is

$$-\lambda^3 + 9\lambda^2 - 21\lambda + 9 \quad (23)$$

which gives eigenvalues $3, 3 + \sqrt{6}, 3 - \sqrt{6}$ which are all positive since $\sqrt{6} < 3$

(c)

consider $\phi_K = \begin{pmatrix} \phi_{K_1} \\ \phi_{K_2} \end{pmatrix}$

the product of any $\phi_K(v_1) \cdot$

$\phi_K(v_2) = K_1(v_1, v_2) + K_2(v_1, v_2)$.

algebraically, any bilinear psd operator $K : V \times V \rightarrow V$ is closed under addition.

$$\langle v, (K_1 + K_2)v \rangle = \langle v, K_1 v \rangle + \langle v, K_2 v \rangle \geq 0 \quad (24)$$

to answer why RBF is PSD, we note that

$$\|x - z\|^2 = \|x\|^2 + \|z\|^2 - 2x \cdot z \quad (25)$$

and thus

$$K(x, z) = \exp(-\gamma\|x\|^2) \exp(-\gamma\|z\|^2) \exp(2\gamma x \cdot z) \quad (26)$$

the product of PSD operators is PSD and thus the final term after being taylor expanded is a sum of scaled product of PSD operators ($x \cdot z$ is PSD)